100 years of

Plant Biology

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#PlantBio2024 | plantbiology.aspb.org

How to make an effective graphical abstract or scientific illustration

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Science Art Enthusiast
2023 Plantae Fellow

Twitter(X): @alicevpierce
Instagram: @alice.pipettes



Pierce et al. 2023 *PSB*

Sonhita Chakraborty, PhD

Postdoc at Umeå Plant Science Centre

Associate Features Editor with *The Plant Cell*

Former editor for Elsevier / Cell Press

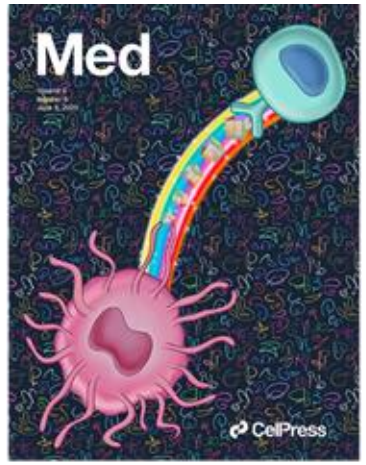
Freelance Science illustrator



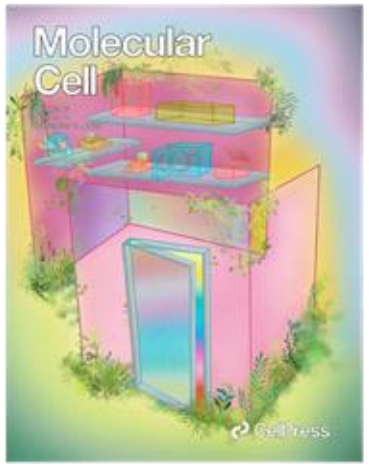
Twitter (X): @qwertess

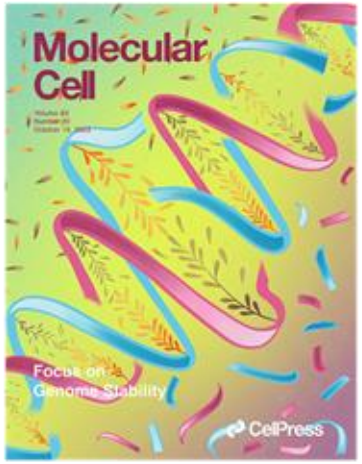
Instagram: @cyberabbit

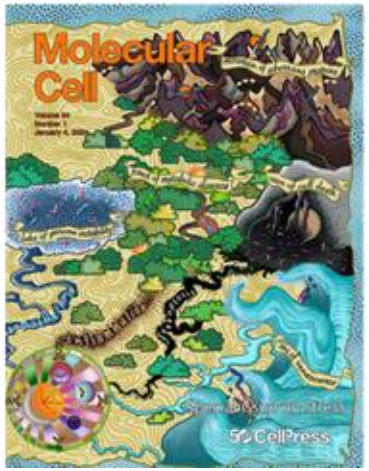










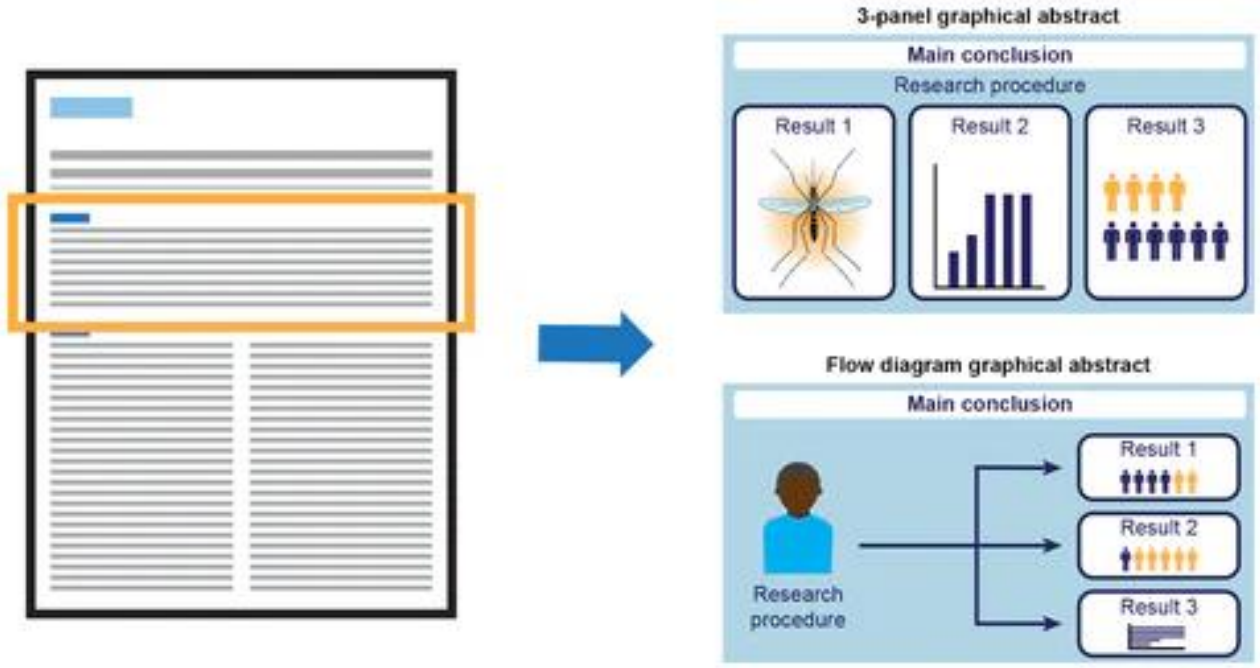


Workshop Outline

- What is a graphical abstract
- Insights from science illustrators (video recordings)
- Graphical abstract fundamentals (videos)
- Paid vs open-source software options
- Exercise → create your own graphical abstract
- Q&A and time to work on your graphical abstract/scientific illustration

What is a Graphical Abstract?

- Visual summary of the main findings and/or key concepts of a research paper



Springer Nature, What is a Graphical Abstract?

Why use a Graphical Abstract?

- Attract your audience
- Spark curiosity
- Accessibility
- Increase impact



Springer Nature, What is a Graphical Abstract?

What makes for a good GA?

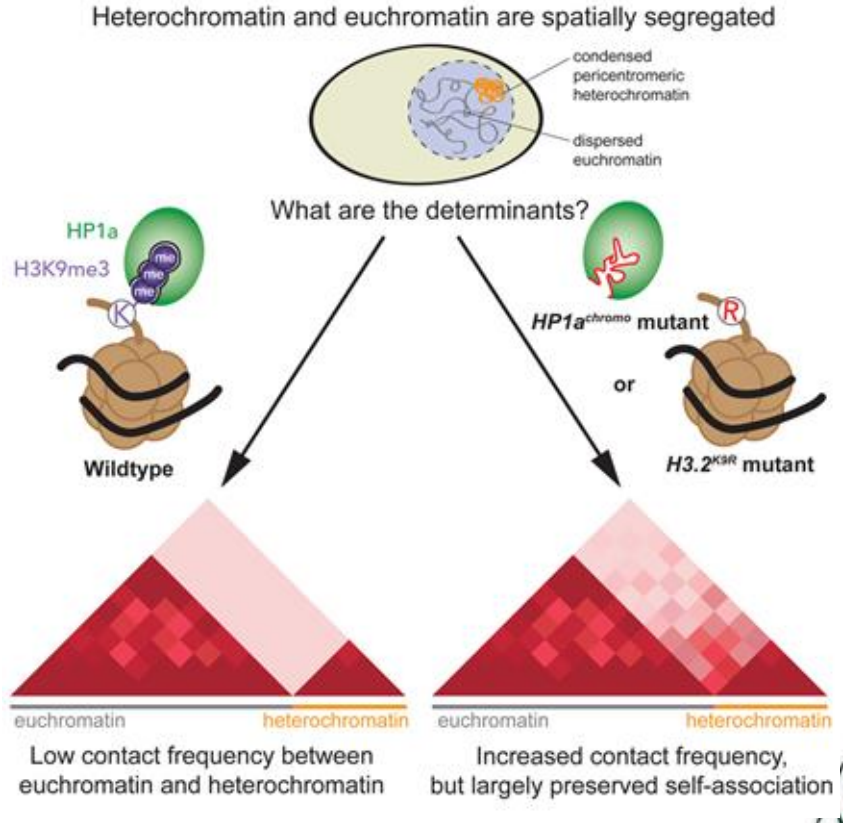
Polleverywhere:

PollEv.com/alicepierce259

- click skip when it prompts to sign in for credit
- answer in one word

What makes for a good GA?

1. Clear and simple
2. Attracts the audience
3. Complements the manuscript
4. Informative and relevant
5. Stimulates curiosity



Stutzman et al., 2024 *Molecular Cell*

Insights from science illustrators

Guest Speaker: Dr. Besiana Sinanaj

Guest Speaker:
Dr. Besiana Sinanaj

Research Assistant, John Innes Centre, UK

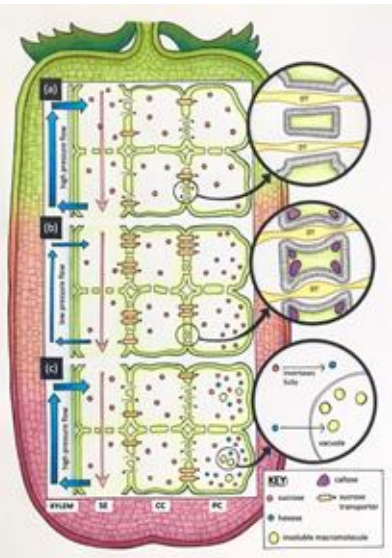
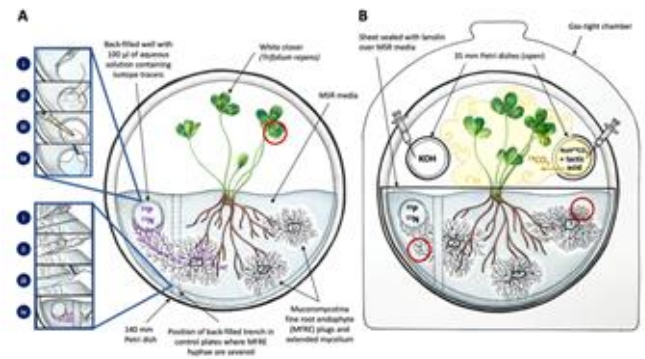
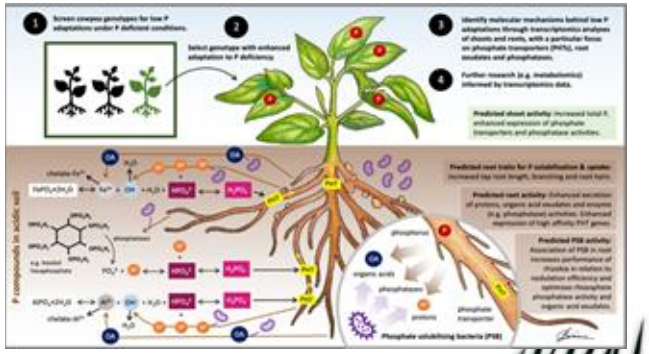
Director & Artist, BESIANA LTD

Interests in plant-soil interactions, mycorrhizal symbiosis, legumes & sustainable agriculture.

Scientific research experience at:

University of Bristol
University of Leeds
Kew Royal Botanic Gardens
University of Sheffield
John Innes Centre



Current Opinion in Plant Biology
Current Biology
New Phytologist

@BesianaX

Fundamentals: things to consider

- Power of the line
- Shape and form
- Textures
- Balance



LearnFree

How to layout your graphical abstract

- Proximity
- Use negative space to yo advantage
- Alignment
- Consider your scientific story: Linear vs circular



LearnFree

Select the best layout for your information:



CATALYZING SCIENCE

Adding color to your graphical abstract

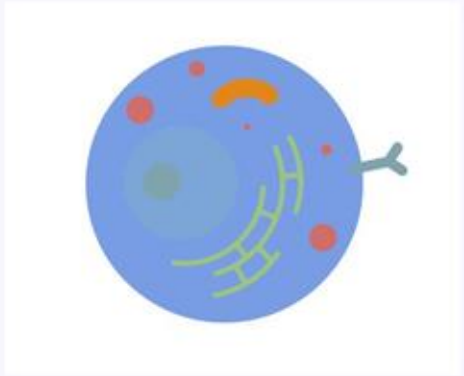
- Color theory and color wheel
- Hue, saturation, value
- Monochromatic, analogous, complementary schemes
- Use colors judiciously to make your ideas pop
- Consider making your GA color blind friendly

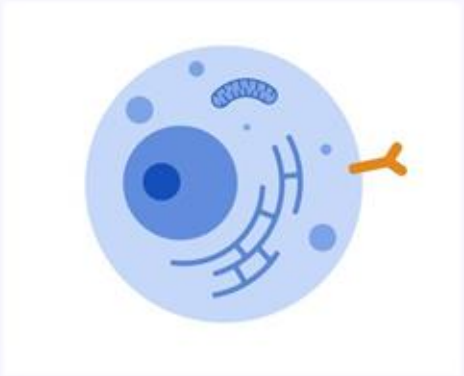


LearnFree

Useful resources: https://color.adobe.com/ https://colorbrewer2.org/

Make sure there is enough contrast between colors:

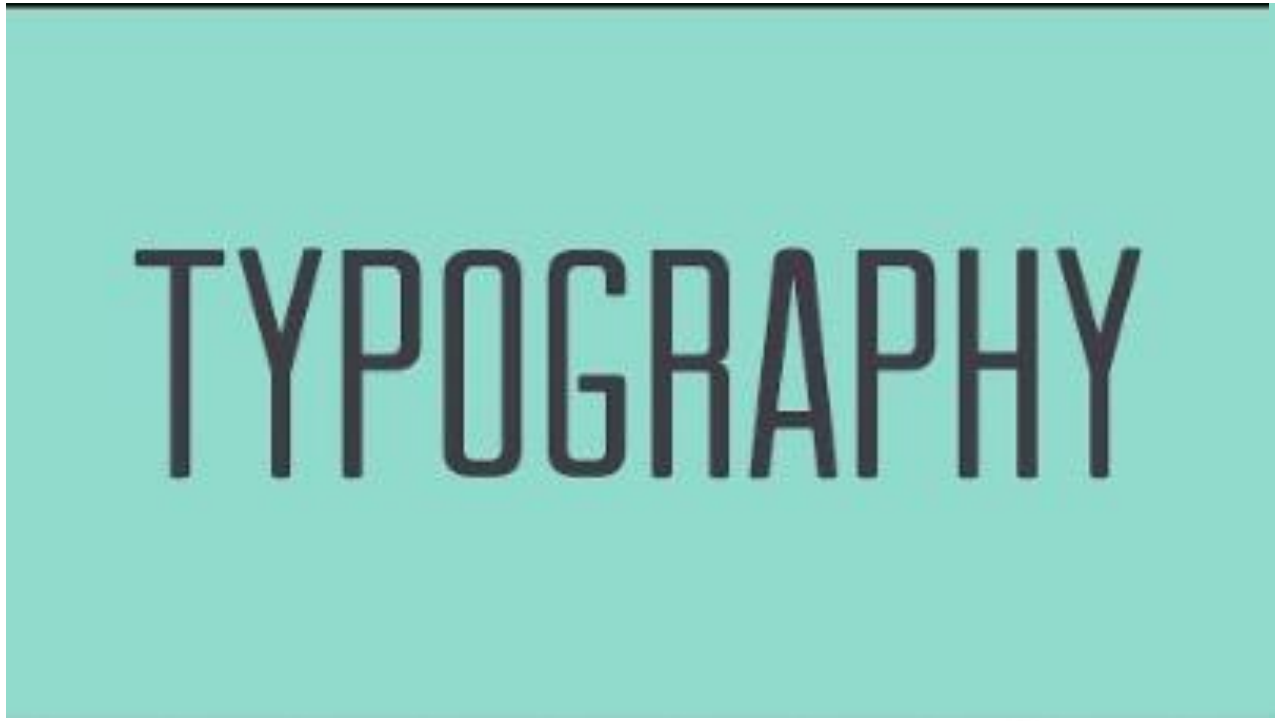




CATALYZING SCIENCE

Typography:

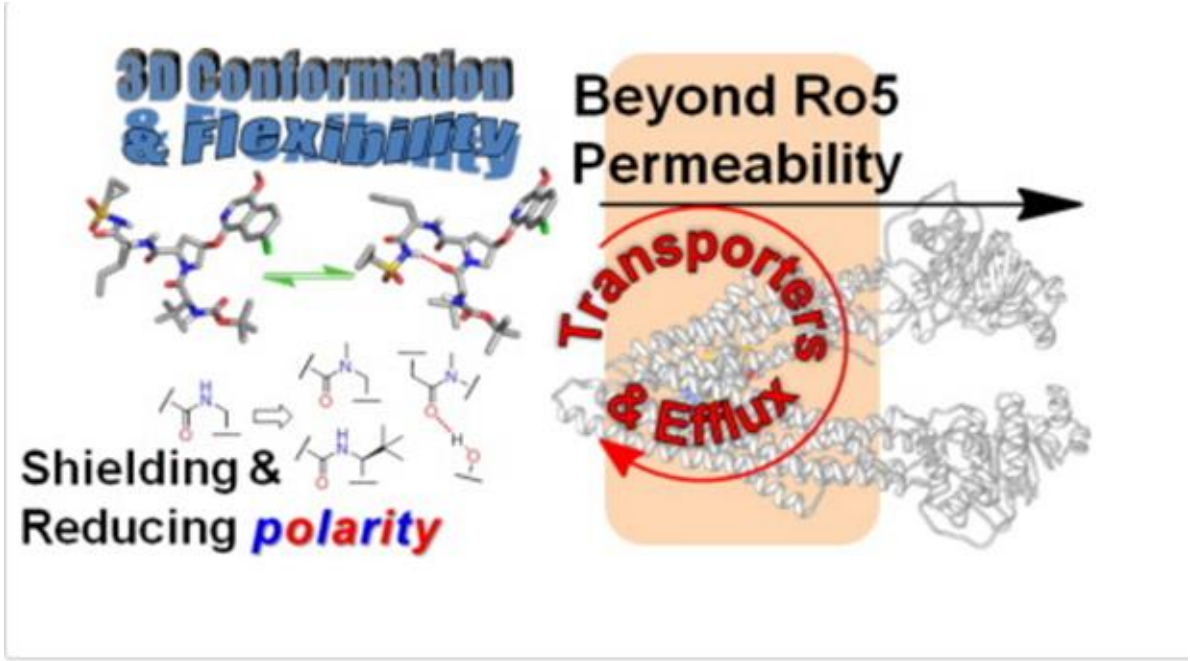
- Sans serifs may be easier to read
- Don’t go overboard with display fonts. Less is more!
- Hierarchy
- Find complementary fonts to create harmony
- Leading, kerning, tracking



LearnFree

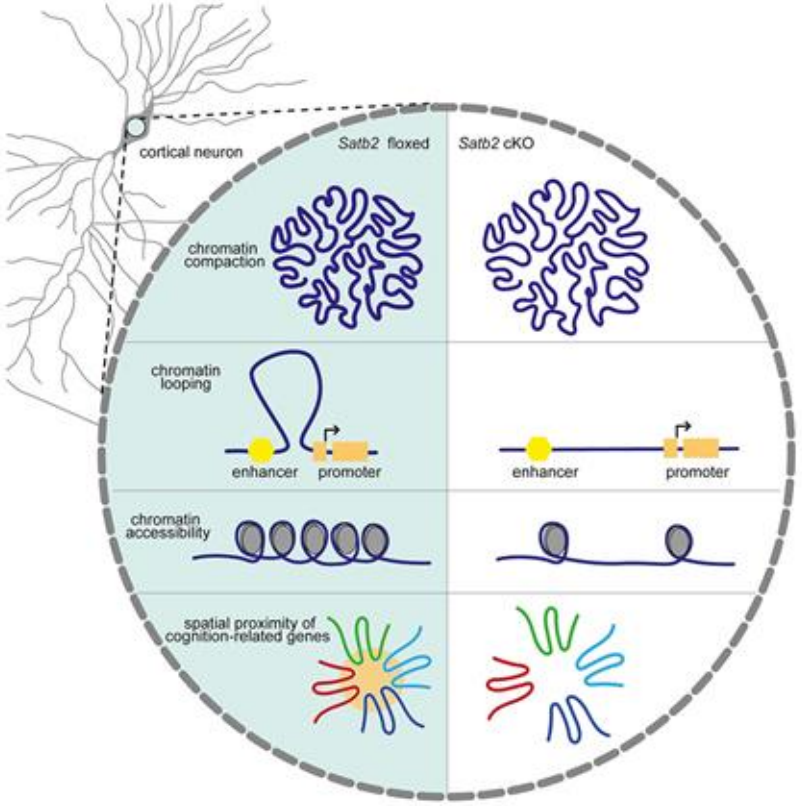
Explore free fonts https://fonts.google.com/

Could be improved



Reddit

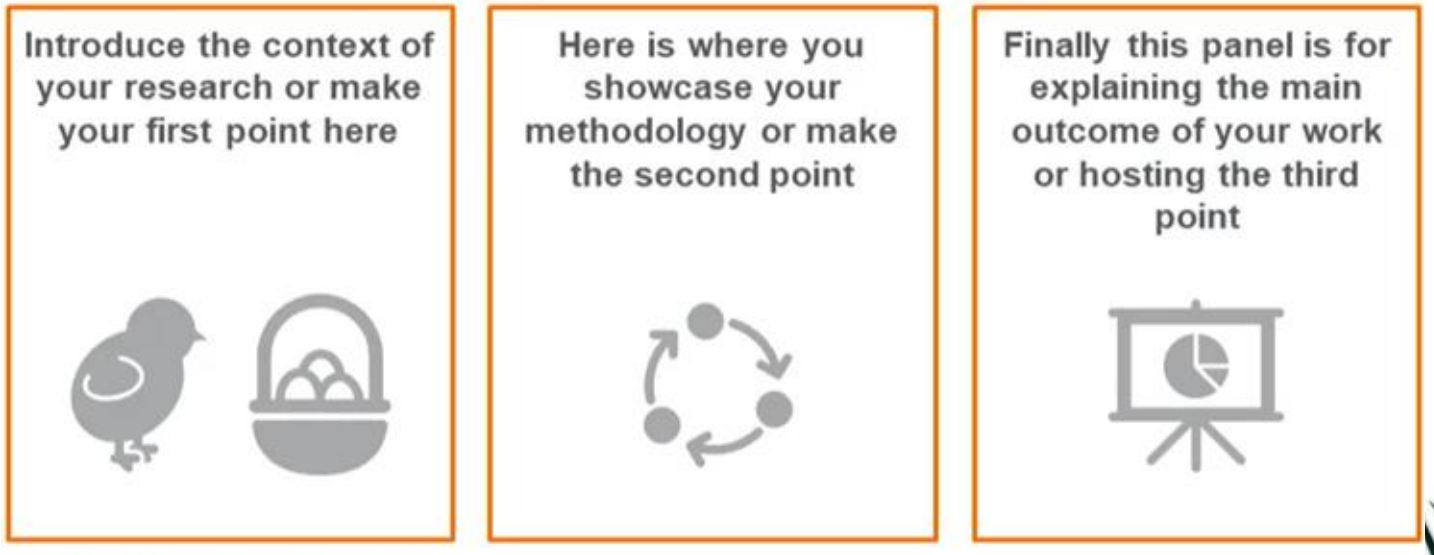
Good



Wahl et al., 2024 *Molecular Cell*

Process from start to finish

1. Brainstorm key message
2. Sketching
3. Iconography
4. Draft(s)
5. Feedback!!
6. Final Illustration



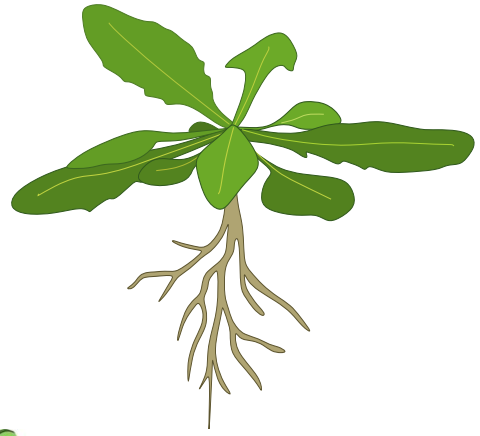
Elsevier, example

Paid vs open source materials

Process	Paid options	Open Source options
Sketching and Layouts	Procreate	Google jam board, FigJam, paper
Iconography	Adobe Illustrator, Biorender	Bioicon
Final Sketch	Adobe Illustrator	Inkscape

Guest Speaker: Dr. Dani Stevens





Genetics and Genomics Ph.D. from UC Davis

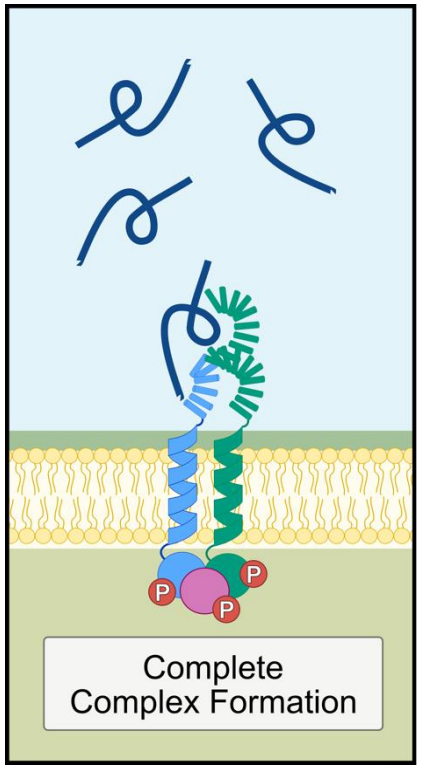
Science Art/Communication Enthusiast

Creator of inkscapeforstem.com

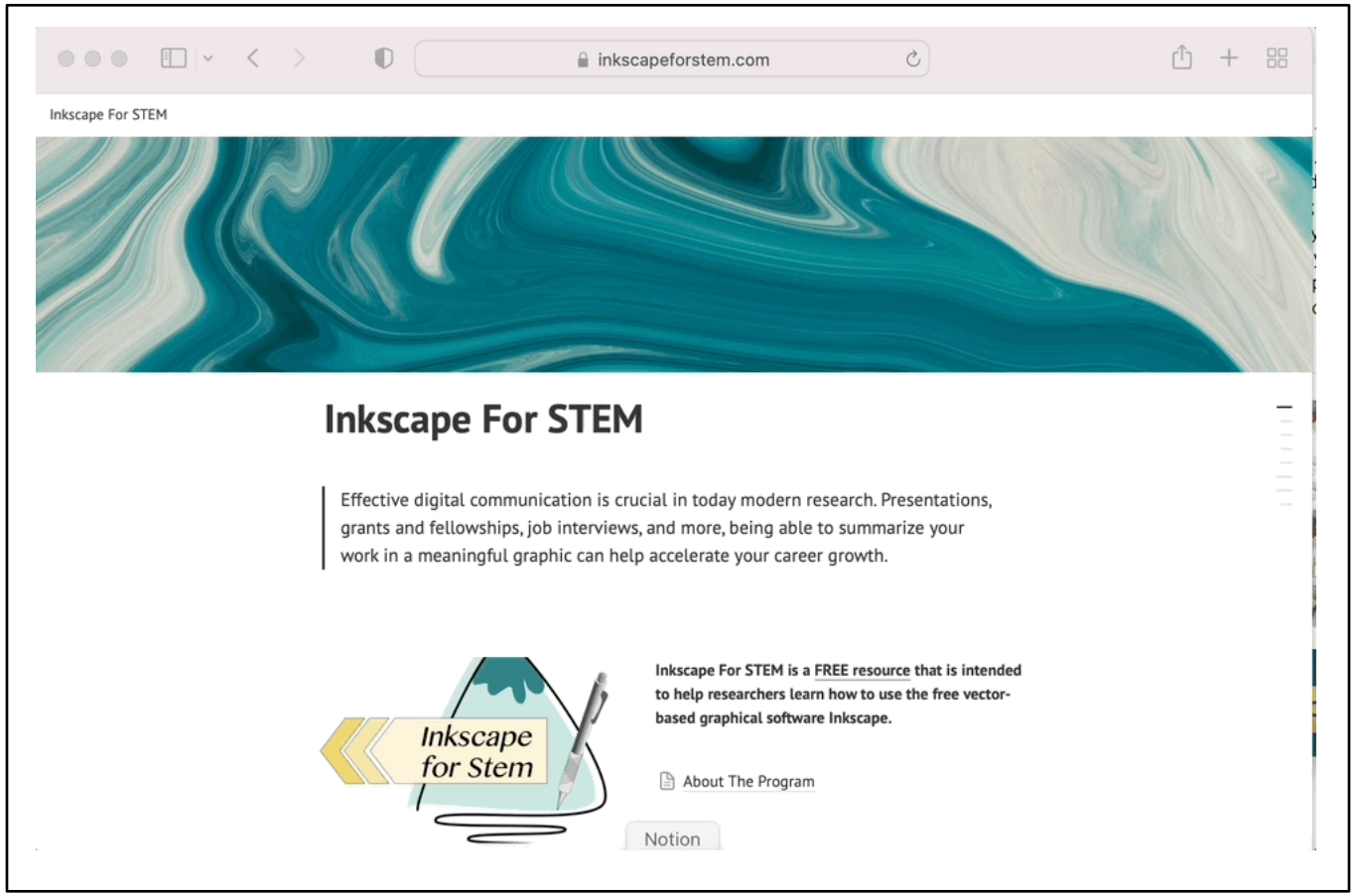
Twitter(X): @Dani_M_Stevens



Plants get sick too.



InkscapeforSTEM is a free web resource



Inkscape is an open-source, free graphic software!

- Our goal is to introduce the software while showing how to make figures without any graphical skills.

Checkout: Inkscapeforstem.com

Walk through exercise

How to make a graphical abstract or scientific illustration

Walk through exercise

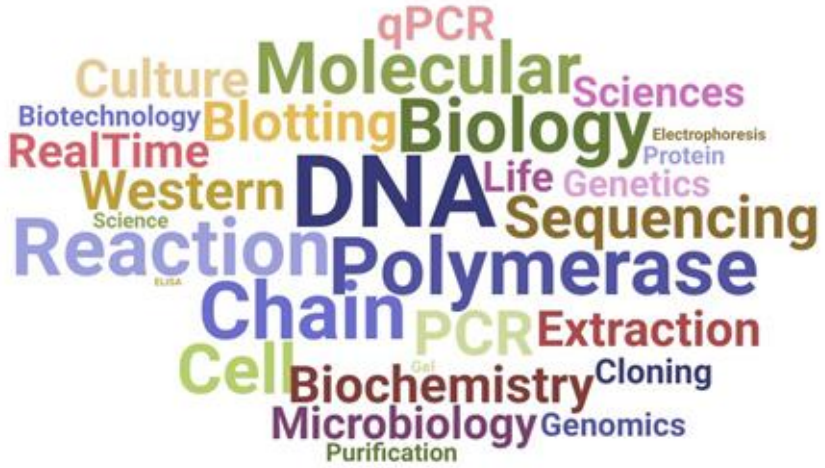
- Introduce yourself to your neighbor!
  - Elevator pitch
    - What do you research?
    - What is the main message you want to illustrate today?

Walk through exercise

- Find material to sketch
- Divide your ‘page’ into 4 parts

Walk through exercise

Part 1: Write down key words associated with the main message (3 min)

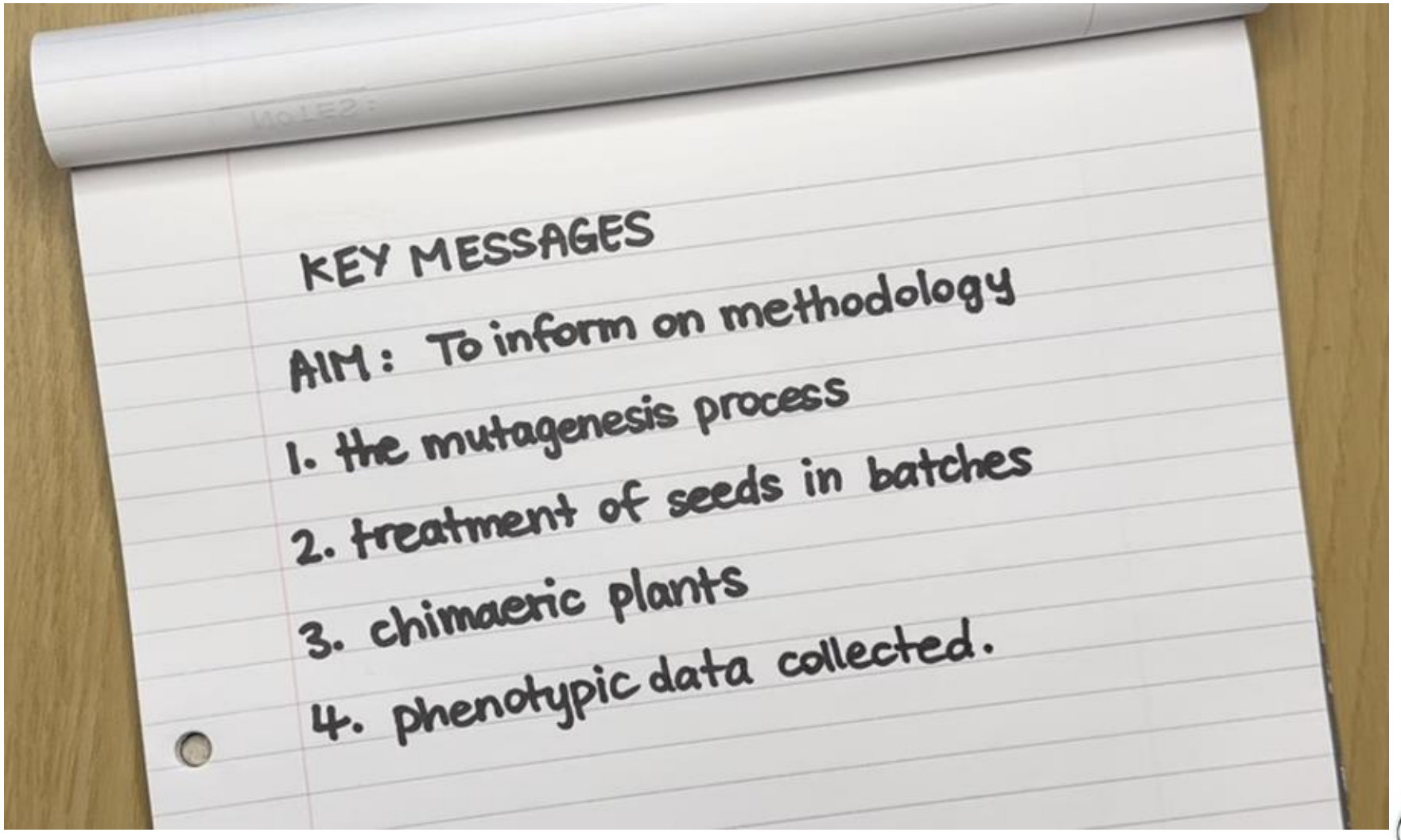


Walk through exercise

Part 1: Write down key words associated with the main message (6 min)

- What key words did you use? (polleverywhere)

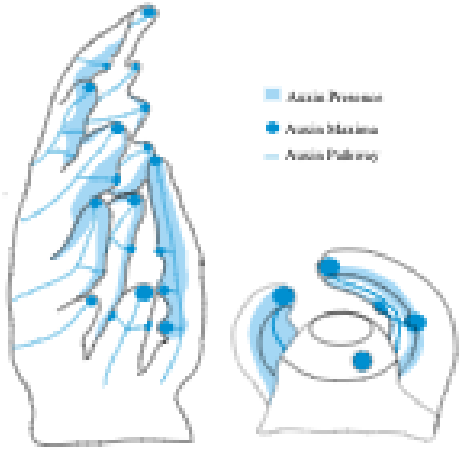
PollEv.com/alicepierce259

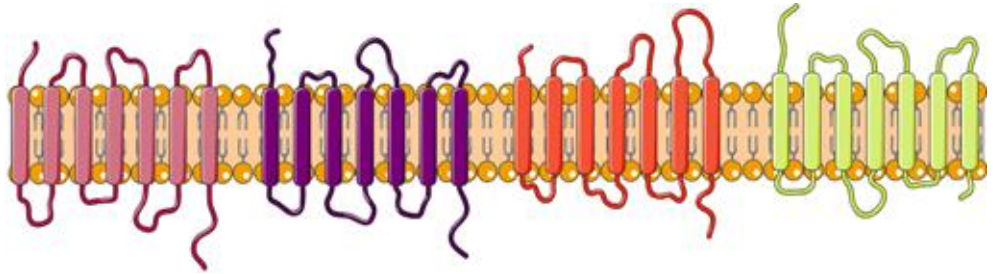

KEY MESSAGES
AIM: To inform on methodology
1. the mutagenesis process
2. treatment of seeds in batches
3. chimaeric plants
4. phenotypic data collected.

Walk through exercise

Part 2: Find and draw key visual elements (8 min)





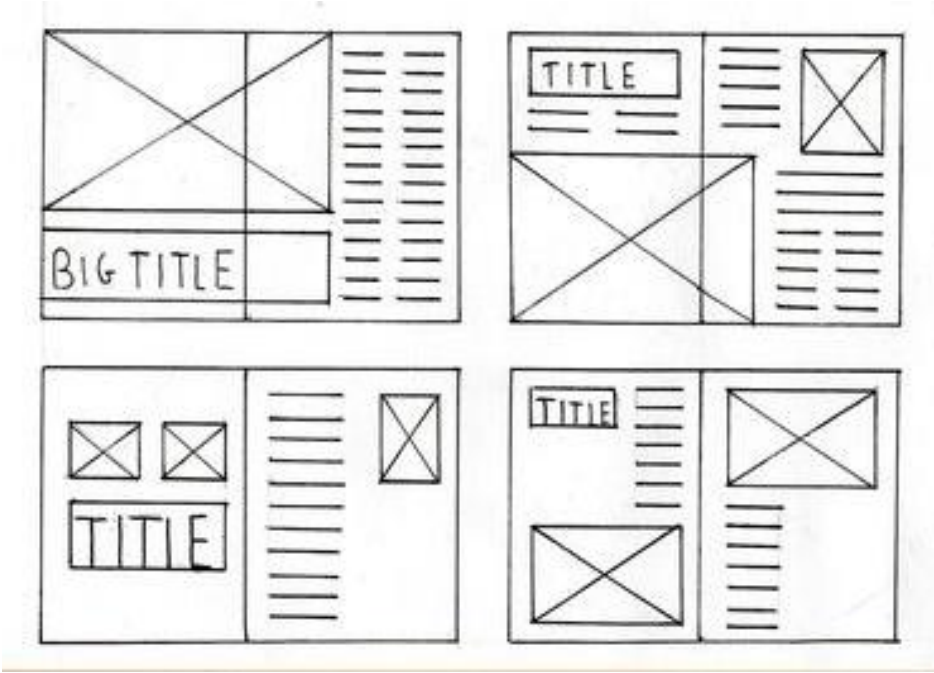




bioicons.com

Walk through exercise

Part 3: Lay out the composition of your GA (12 min)



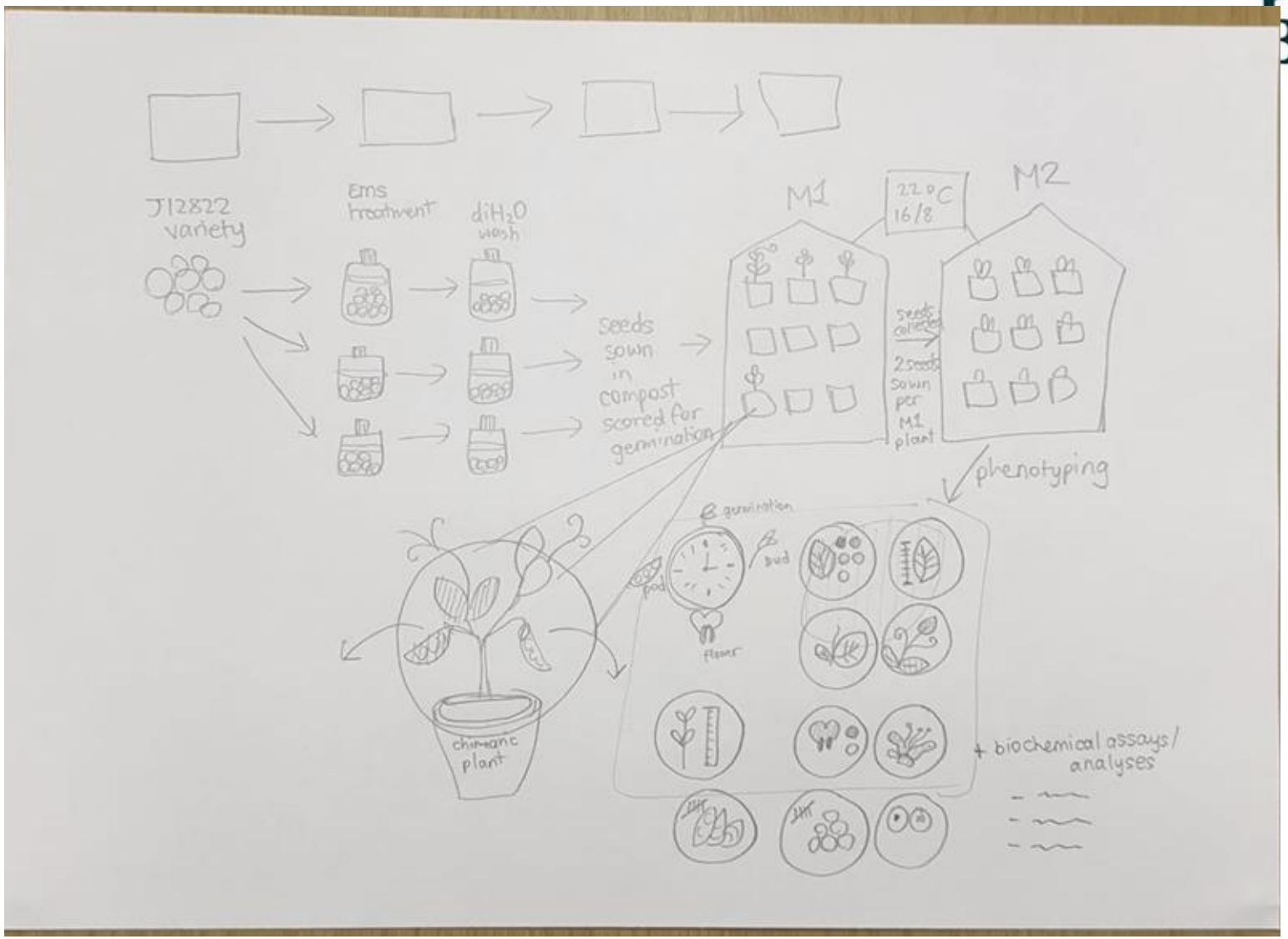
Select the best layout for your information:



CATALYZING SCIENCE

Walk through exercise

Part 4: Incorporate parts 1-3 into a rough sketch (15 min)


J12822 variety
Ems treatment
diH2O wash
Seeds sown in compost scored for germination
M1
22°C 16/8
M2
Seeds collected
2 seeds sown per M1 plant
phenotyping
germination
bud
flower
chimeric plant
+ biochemical assays/ analyses

Walk through exercise

Part 5: Swap your sketch with your neighbor to the RIGHT

- (2 minutes) Analyze sketch
- (6 minutes) Give constructive feedback to each other
  - What did they do well?
  - How did they incorporate insights learned from the workshop? (layout, color, typography)
  - What can they improve on?

Walk through exercise

Part 6: Incorporate feedback into your ‘final’ sketch (10 minutes)

Walk through exercise

Part 7: Swap your sketch with your neighbor to the LEFT

- (2 minutes) Analyze sketch
- (6 minutes) Give constructive feedback to each other
  - What did they do well?
  - How did they incorporate insights learned from the workshop? (layout, color, typography)
  - What can they improve on?

Walk through exercise

Part 8: Congrats! You made a graphical abstract draft



Thank you!

Please give us feedback on this workshop!

https://tinyurl.com/graphicalabstractworkshop

Connect with us!

Alice:

E-mail: avpierce@ucdavis.edu

Twitter(X): @alicevpierce

Instagram: @alice.pipettes

Sonhita:

E-mail: sonhita.chakraborty@slu.se

Twitter (X): @qwertess

Instagram: @cyberabbit

Q&A

Time to talk with Dr. Sinanaj
Work on your draft

Connect with Besiana!

BESIANA company website: www.besiana.co.uk
Linked In: https://uk.linkedin.com/in/besiana-sinanaj-106744109
X: @BesianaX
Instagram: art.by.besiana